

Homework 1

1. Suppose $u \in C^2(\Omega)$ and is radial, i.e.,

$$u = u(r), \forall x \in \Omega \subset \mathbb{R}^n, \text{ where } r = \sqrt{x_1^2 + \cdots + x_n^2}.$$

(i) Show that

$$\Delta u = u_{rr} + \frac{n-1}{r}u_r, \quad \forall x \in \Omega.$$

(ii) Show that if

$$u(r) = \begin{cases} \ln r, & n = 2, \\ \frac{1}{r^{n-2}}, & n \geq 3, \end{cases}$$

then u is harmonic in $\mathbb{R}^n \setminus \{0\}$.

2. Prove that if “ $Lu \geq 0$ ” is replaced by “ $Lu = 0$ ” in the weak maximum principle with $c \leq 0$, then

$$\max_{\bar{\Omega}} |u| = \max_{\partial\Omega} |u|. \quad (1)$$

3. Assume that Ω is a bounded domain in $\mathbb{R}^n$.

(i) Prove that the following Dirichlet boundary problem has at most one solution in $C^2(\Omega) \cap C^0(\bar{\Omega})$:

$$\begin{cases} \Delta u + f(u) = 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases} \quad (2)$$

where $f \in C^1(\mathbb{R})$ satisfies $f'(u) < 0$ for all $u \in \mathbb{R}$.

(ii) Give an example to show that Part (i) may not hold without the assumption “ $f'(u) \leq 0$ ”.

4. Consider the partial differential operator

$$Lu = \sum_{i,j=1}^n a_{ij}(x) \frac{\partial^2 u}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i(x) \frac{\partial u}{\partial x_i} + c(x)u, \quad x \in \Omega, \quad (3)$$

where Ω is a bounded domain in $\mathbb{R}^n$, L is strictly elliptic on Ω , a_{ij} and b_i are bounded on Ω .

(i) For $n = 1$, give an example of L and u such that

$$\begin{cases} Lu \geq 0 & \text{in } \Omega, \\ u \leq 0 & \text{on } \partial\Omega, \end{cases} \quad (4)$$

yet, “ $u \leq 0$ on Ω ” is not true. (This means weak maximum principle does not hold without sign condition on $c(x)$.)

(ii) Yet, we can replace the sign condition on $c(x)$ by another condition to have “(4) implies $u \leq 0$ on Ω ”: Suppose $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ satisfies (4); assume there is a $v \in C^2(\Omega) \cap C^0(\overline{\Omega})$ such that

$$\begin{cases} Lv \leq 0 & \text{in } \Omega, \\ v > 0 & \text{on } \overline{\Omega}, \\ \text{all } v_{x_i} & \text{are bounded on } \Omega. \end{cases} \quad (5)$$

Prove that $u \leq 0$ on Ω .

Hint: Let $w = \frac{u}{v}$, which implies $L(vw) \geq 0$ in Ω . Then, find another linear operator $\tilde{L}$ s.t. $\tilde{L}w \geq 0$ in Ω , where “ c ” in $\tilde{L}$ has the “good sign”.

5. Let B_1 be the closed unit ball in $\mathbb{R}^n$, centered at 0. Suppose $u \in C^2(B_1 \setminus \{0\})$ is harmonic in $B_1 \setminus \{0\}$ (u may be singular at 0). Suppose

$$u(x) = \begin{cases} o(\ln r) & \text{as } r = |x| \rightarrow 0+, \quad n = 2, \\ o\left(\frac{1}{r^{n-2}}\right) & \text{as } r = |x| \rightarrow 0+, \quad n \geq 3. \end{cases}$$

Prove that

$$\sup_{B_1 \setminus \{0\}} u = \sup_{\partial B_1} u, \quad \inf_{B_1 \setminus \{0\}} u = \inf_{\partial B_1} u.$$

Hint: For $n = 2$, consider $u(x) \pm \epsilon \ln r$ for the sup and inf identities, respectively. For $n \geq 3$, consider $u(x) \mp \frac{\epsilon}{r^{n-2}}$ for the sup and inf identities, respectively.

6. For $x \in B_1$, define $v(x) = |x|^{2-n}u(\frac{x}{|x|^2})$, where $\Delta u = 0$ in $\overline{B_1^c}$. Prove that $\Delta v = 0$ in $B_1 \setminus \{0\}$. (v is called the Kelvin transform of u .)